

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0098

Indian Lake 138 kV
December 20, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0098. The Interconnection Customer proposes a 176.4 MW wind facility interconnecting at the Indian Lake 138 kV substation in the Virginia area. The request was received on January 21, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 71 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,883,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pleasant View / Claytonia 138 kV
Contingency: Loudon / Thomaston 345 kV line
Loading: 122.7% (Rating: 221 MVA)
DFAX: 65.2%
Pre-project loading: 70.7%
2. Facility: Indian Lake / Pleasant View 138 kV
Contingency: Thomaston / Loudon 345 kV
Loading: 111.2% (Rating: 288 MVA)
DFAX: 65.2%
Pre-project loading: 71.4%
3. Facility: Wheeler / Pasco Cnty Solid Waste Resource Recovery 138 kV
Contingency: Loudon / Thomaston 345 kV
Loading: 100.6% (Rating: 231 MVA)
DFAX: 11.7%
Pre-project loading: 91.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pleasant View / Claytonia 138 kV	Rebuild line with higher-capacity conductor	\$7,875,000	100% (\$7,875,000)
Indian Lake / Pleasant View 138 kV	Rebuild line with higher-capacity conductor	\$11,979,000	100% (\$11,979,000)
Wheeler / Pasco Cnty Solid Waste Resource Recovery 138 kV	Reconductor line	\$1,745,000	59% (\$1,029,000)
	Total allocated to project		\$20,883,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.